⚠ IMPORTANT — Support Scope

3DPTronics supports the VirtuaTilt controller hardware only. This guide is a *compendium* to help you connect your VirtuaTilt to third-party pinball software (VPX, Future Pinball + BAM, DOF/DOFLinx, PinEvent, PinUP Popper, etc.). We provide them as a courtesy to save you the hunt across multiple sites. **The information is best-effort** and kept reasonably current, but is **not** official documentation of those projects. **We do not provide official support for any third-party software referenced here** — for software-specific issues, refer to each project's own docs and community channels.

☑ Officially supported

Component	Coverage
🎮 **VirtuaTilt controller**	Wired and Wireless editions, all hardware components
💾 **Pinscape Pico firmware**	Updates, OTA flashing, recovery via Updater Tool
📋 **Bundled profiles**	All BT and USB profiles shipped in the redist
⚙ **GUIConfigTool**	Configuration, calibration, profile loading, output testing
🔧 **Wireless Kit**	Wireless hardware kit for existing VirtuaTilt owners
📦 **Accessories**	Anything shipped directly by 3DPTronics

❌ Not supported

Category	Examples	Where to get help
🎲 **VP Game engines**	VPX, Visual Pinball, Future Pinball, BAM	VPForums · VP Universe
🕹 **VP titles**	Pinball FX / FX2 / FX3 / VR / M, etc.	Zen Studios support
💥 **Feedback frameworks**	DOF (Direct Output Framework), DOFLinx	DOF on GitHub · DOFLinx wiki
🎨 **DMD / extras**	DMDExt, PinEvent V2, PuP-Packs / PUP-Stream	VP Universe
🚀 **Frontends**	PinUP Popper, Baller Installer, etc.	Popper forum
🎬 **Emulators / ROMs**	MAME, pinball ROMs	MAMEdev
💻 **System-level**	Windows, GPU/audio drivers, USB controllers	Microsoft / vendor support

🌐 Community help for software

Project	Community
Visual Pinball X	vpforums.org · vpuniverse.com
Future Pinball + BAM	VPU Future Pinball forum
Future Pinball + BAM AIO	VPU file 14807 — FP+BAM Essentials AIO
DOF	github.com/DirectOutput/DirectOutput
DOFLinx	doflinx.github.io/docs
Pinball FX family	Zen Studios Support
PinUP Popper	VPU PinUP Popper forum
MAME	mamedev.org

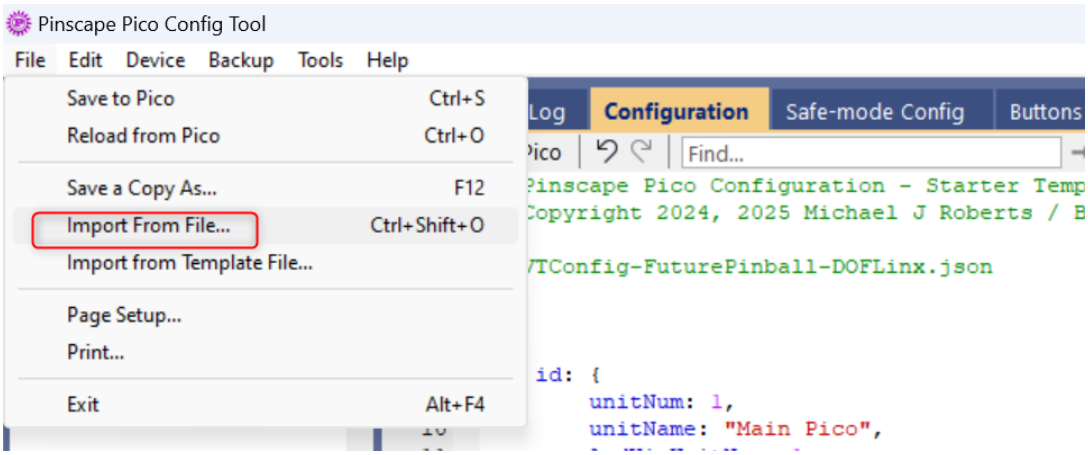
💡 **Why this matters:** Our team is small and our expertise is the VirtuaTilt hardware — we can't debug VPX scripts, flaky BAM installs, or broken DOF chains. Those communities have their own experts. The **AI chatbot on the wiki** can help connect VirtuaTilt with most of the software above; for deeper or vendor-specific issues, reach out to the appropriate community first.

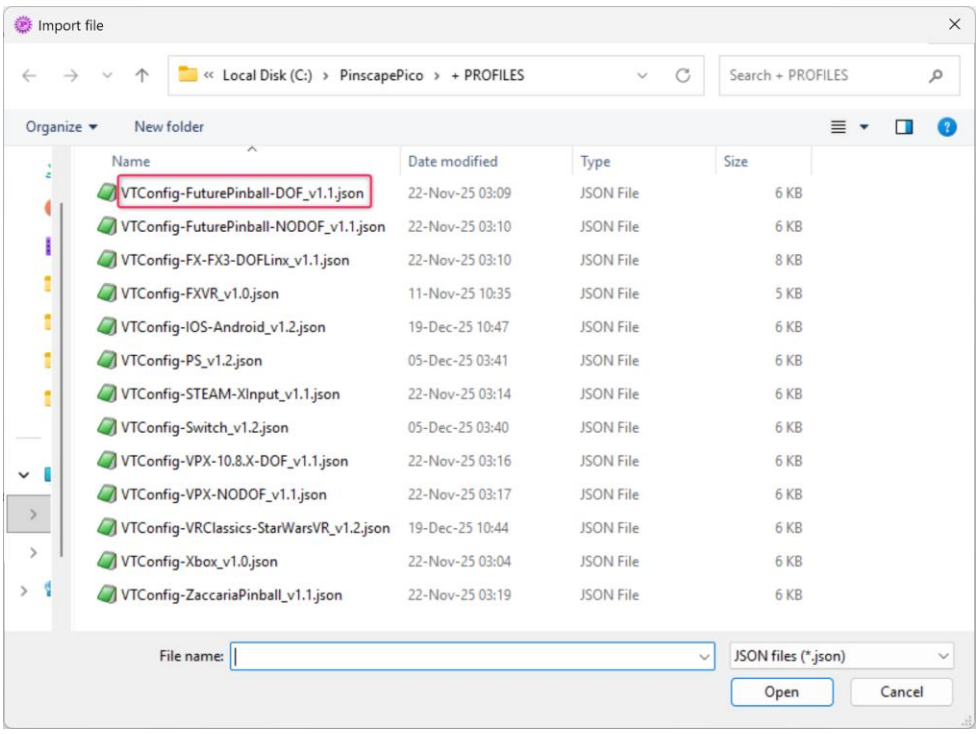
VIRTUATILT – FUTURE PINBALL SETUP

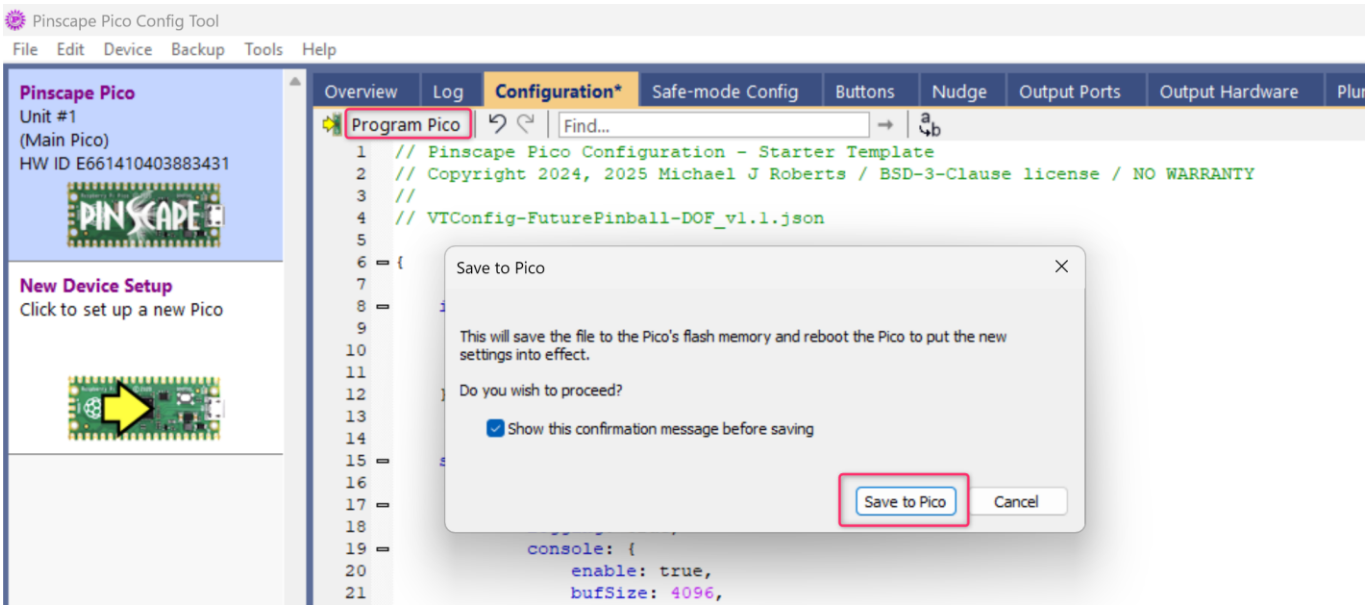
Following some basic instructions to have your new VirtuaTilt up and running with Future Pinball.

PREREQUISITES:

- DirectOutput Framework (**DOF**) already installed and working (refer to **VirtuaTilt-RP2040_VPXSetup_v1.x** document)
- DOF **NOT** running
- Windows System Language and Keyboard set to **English (MANDATORY)**
- **VTConfig-FuturePinball-DOF_v1.X.json** profile loaded in **Pinscape Pico**






Pinscape Pico Config Tool
File Edit Device Backup Tools Help
Pinscape Pico
Unit #1
(Main Pico)
HW ID E661410403883431
New Device Setup
Click to set up a new Pico
Overview Log Configuration* Safe-mode Config Buttons Nudge Output Ports Output Hardware
Program Pico
Find...
// Pinscape Pico Configuration - Starter Template
// Copyright 2024, 2025 Michael J Roberts / BSD-3-Clause license / NO WARRANTY
//
// VTConfig-FuturePinball-DOF_v1.1.json
Save to Pico
This will save the file to the Pico's flash memory and reboot the Pico to put the new settings into effect.
Do you wish to proceed?
Show this confirmation message before saving
Save to Pico
Cancel
console: {
enable: true,
bufSize: 4096,

Future Pinball Installation

Download the **FP and BAM Essentials AIO package** here:

https://vpuniverse.com/files/file/14807-future-pinball-and-bam-essentials-all-in-one-complete/

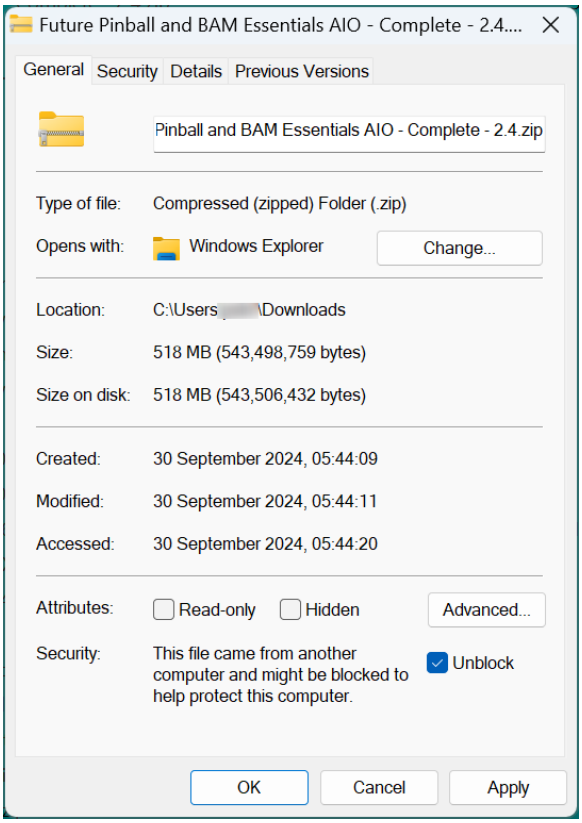
The FP and BAM Essentials AIO package includes a complete install of Future Pinball, the latest BAM update, the latest BAM Open-VR update, and a collection of essential files. This covers desktop, cabinet, and VR players.

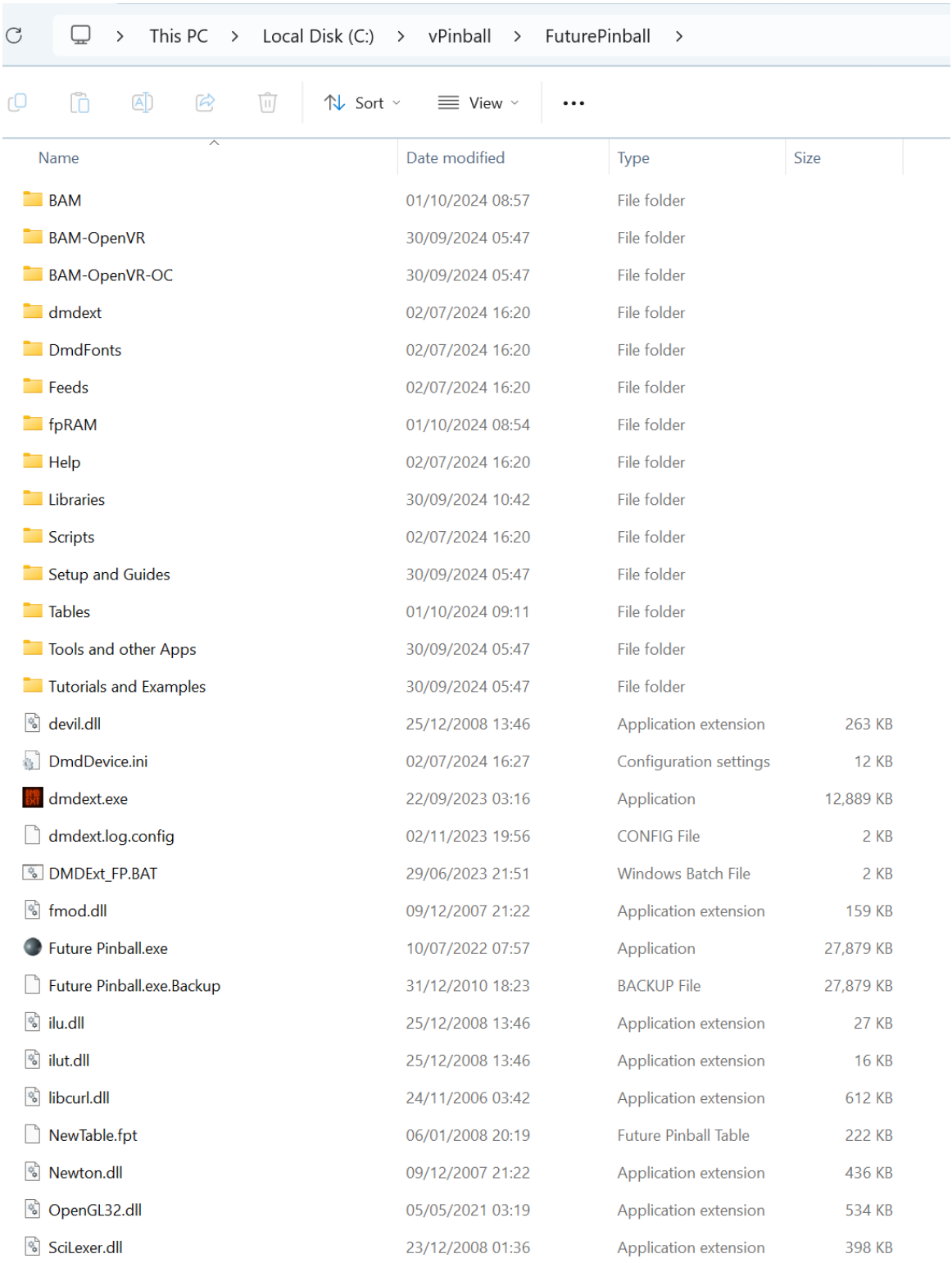
Installing this package makes setting up Future Pinball and BAM easy at any time with everything you need, ready to go in one folder.

This can be used for a new install or an update. If you already have a working Future Pinball and BAM setup, then this package will get you updated with everything you need to play the latest and greatest Future Pinball tables!

No need to use installers or download files and guides from different locations. No need to patch files. It's all here ready to go in one zip file!

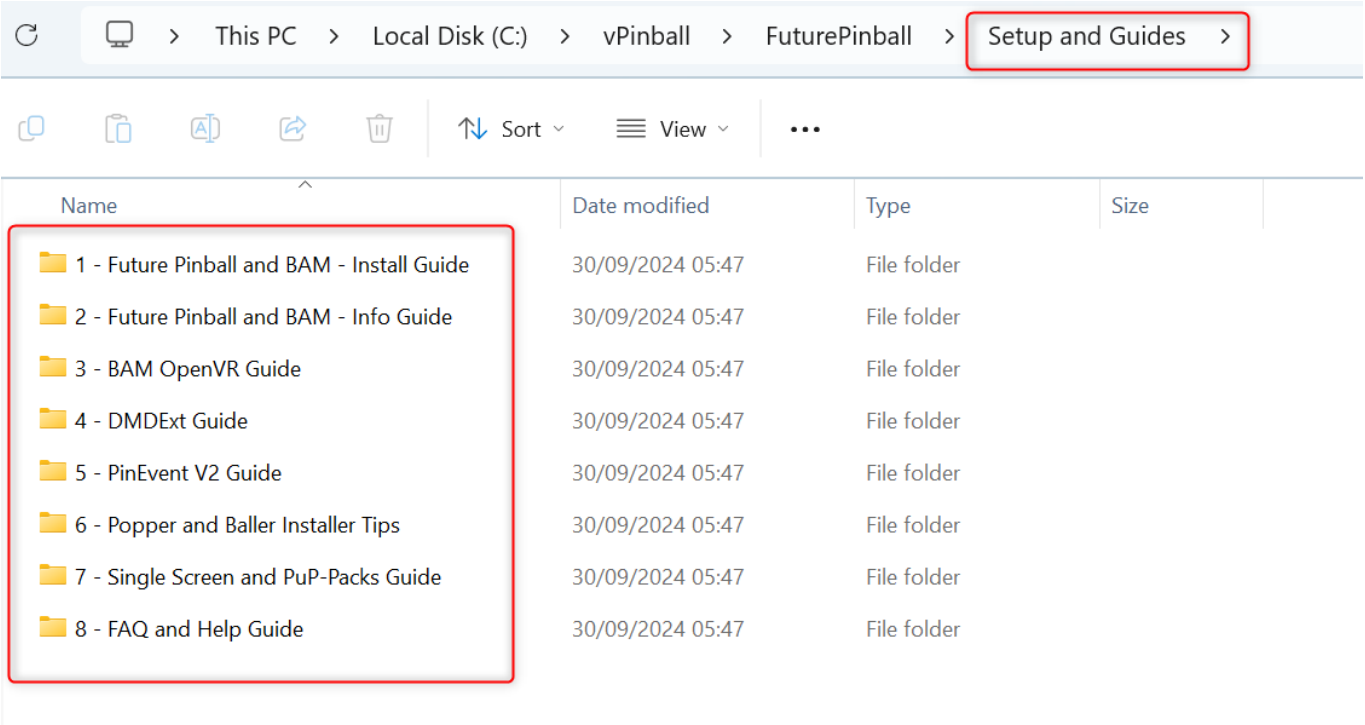
Unblock zipfile and extract it:



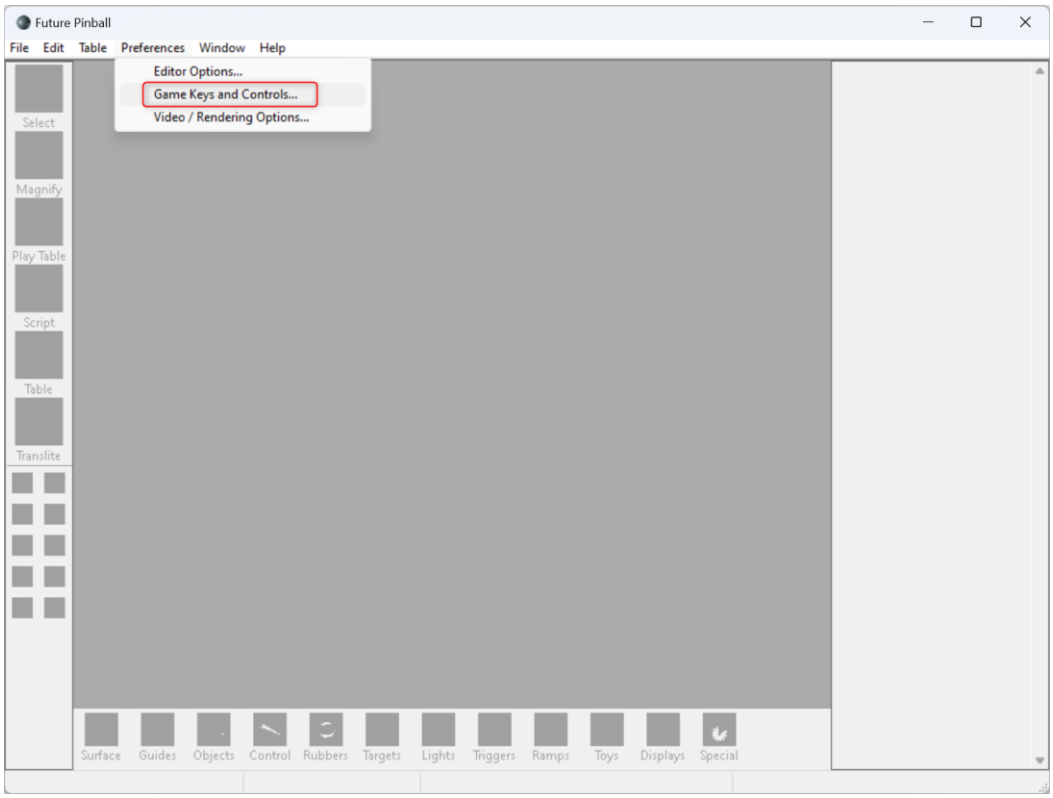


Name	Date modified	Type	Size
BAM	01/10/2024 08:57	File folder	
BAM-OpenVR	30/09/2024 05:47	File folder	
BAM-OpenVR-OC	30/09/2024 05:47	File folder	
dmdext	02/07/2024 16:20	File folder	
DmdFonts	02/07/2024 16:20	File folder	
Feeds	02/07/2024 16:20	File folder	
fpRAM	01/10/2024 08:54	File folder	
Help	02/07/2024 16:20	File folder	
Libraries	30/09/2024 10:42	File folder	
Scripts	02/07/2024 16:20	File folder	
Setup and Guides	30/09/2024 05:47	File folder	
Tables	01/10/2024 09:11	File folder	
Tools and other Apps	30/09/2024 05:47	File folder	
Tutorials and Examples	30/09/2024 05:47	File folder	
devil.dll	25/12/2008 13:46	Application extension	263 KB
DmdDevice.ini	02/07/2024 16:27	Configuration settings	12 KB
dmdext.exe	22/09/2023 03:16	Application	12,889 KB
dmdext.log.config	02/11/2023 19:56	CONFIG File	2 KB
DMDExt_FP.BAT	29/06/2023 21:51	Windows Batch File	2 KB
fmod.dll	09/12/2007 21:22	Application extension	159 KB
Future Pinball.exe	10/07/2022 07:57	Application	27,879 KB
Future Pinball.exe.Backup	31/12/2010 18:23	BACKUP File	27,879 KB
ilu.dll	25/12/2008 13:46	Application extension	27 KB
ilut.dll	25/12/2008 13:46	Application extension	16 KB
libcurl.dll	24/11/2006 03:42	Application extension	612 KB
NewTable.fpt	06/01/2008 20:19	Future Pinball Table	222 KB
Newton.dll	09/12/2007 21:22	Application extension	436 KB
OpenGL32.dll	05/05/2021 03:19	Application extension	534 KB
SciLexer.dll	23/12/2008 01:36	Application extension	398 KB

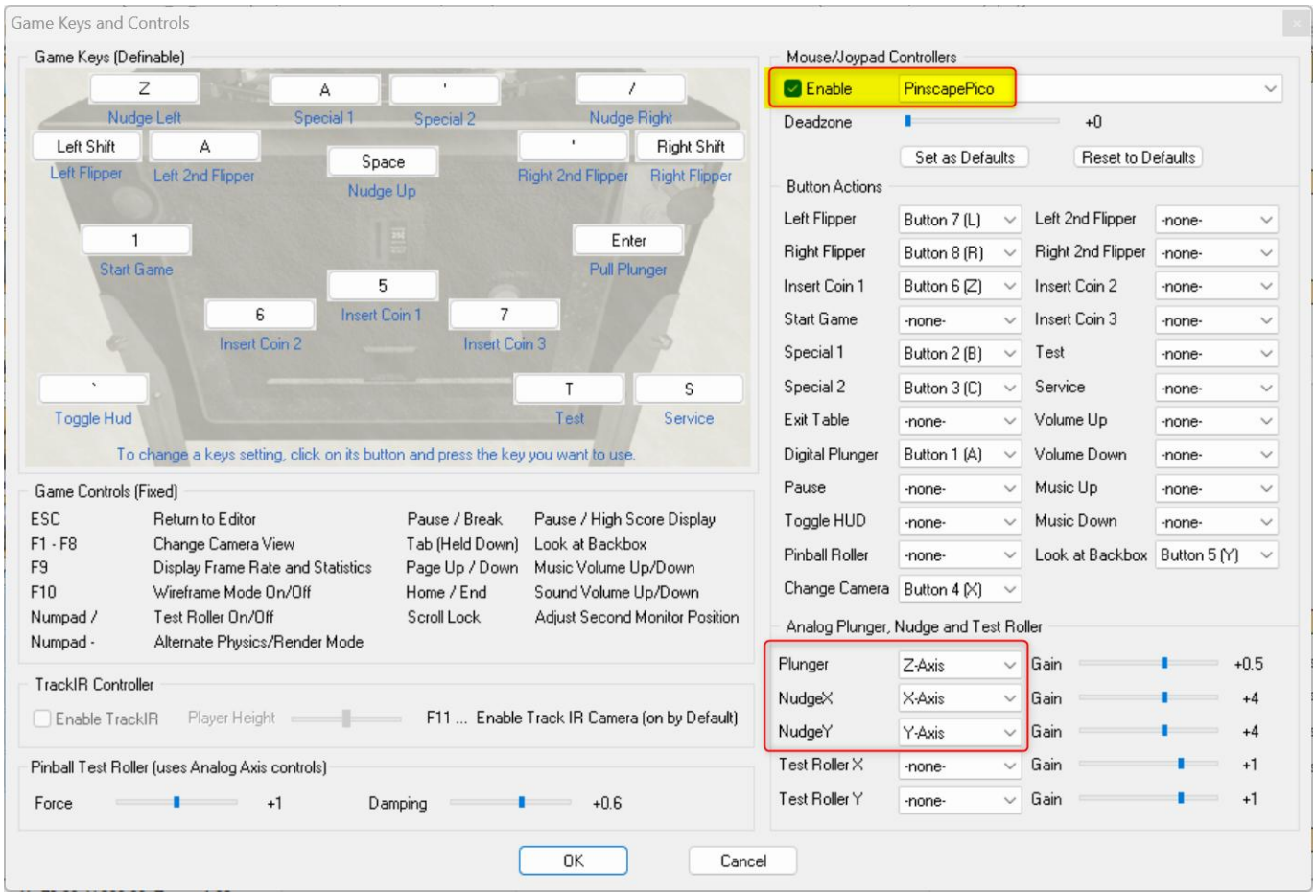
Follow official Install guide available in the extracted folder to set up BAM for your cabinet configuration **(STEPS 1-13)**:



After you configured FP from the above guides, launch your **Start FP – xxxxx.bat** and edit **Game/Keys and Controls** preferences:

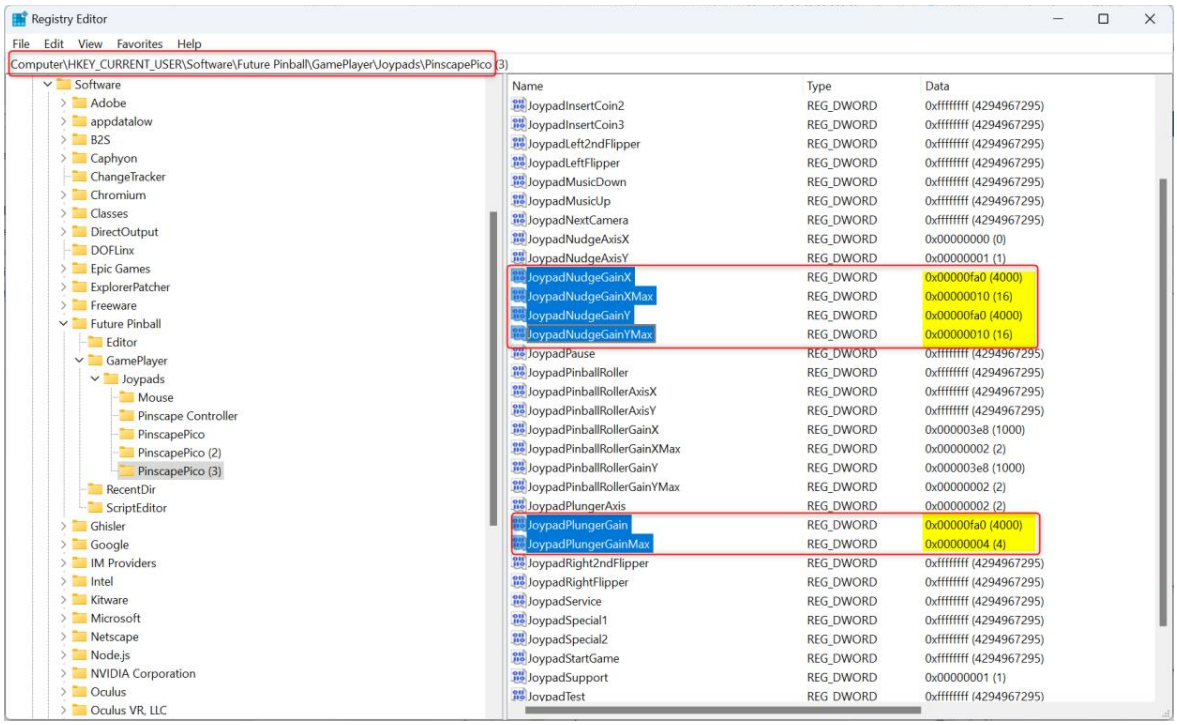


Select and Enable **PinscapePico Controller:**

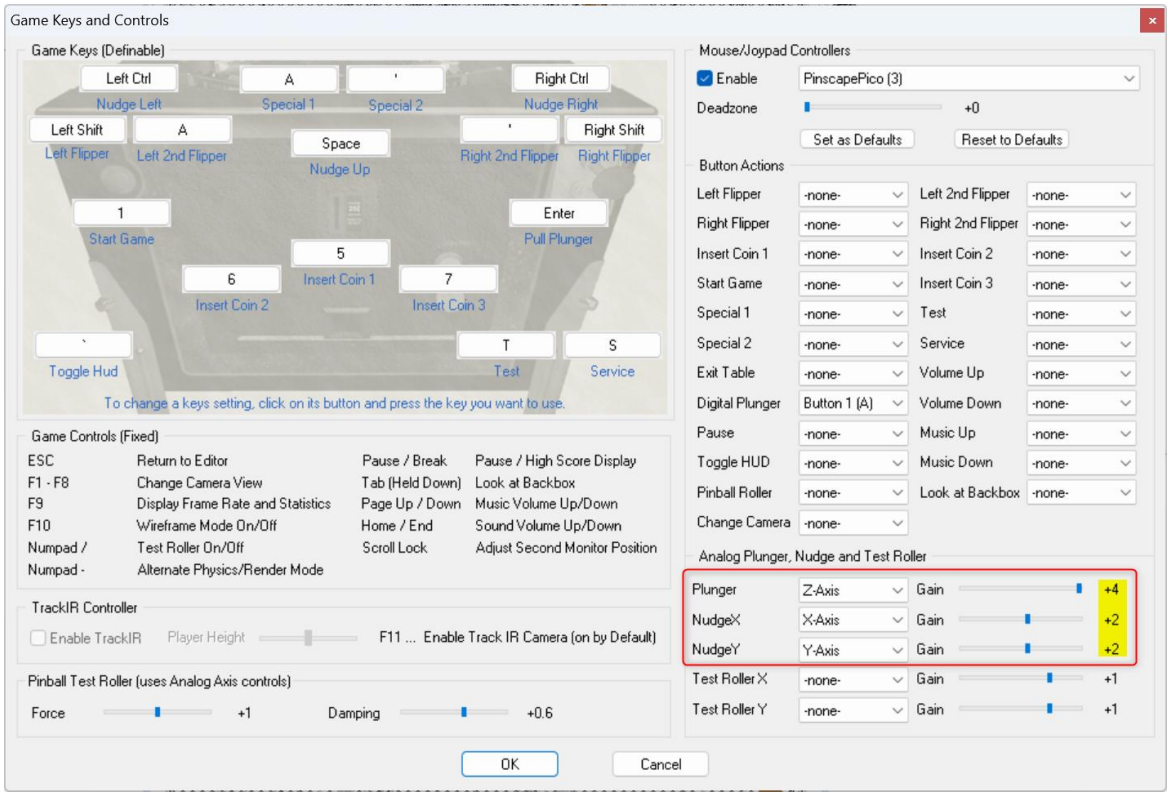


Accelerometer/Analog Plunger configuration

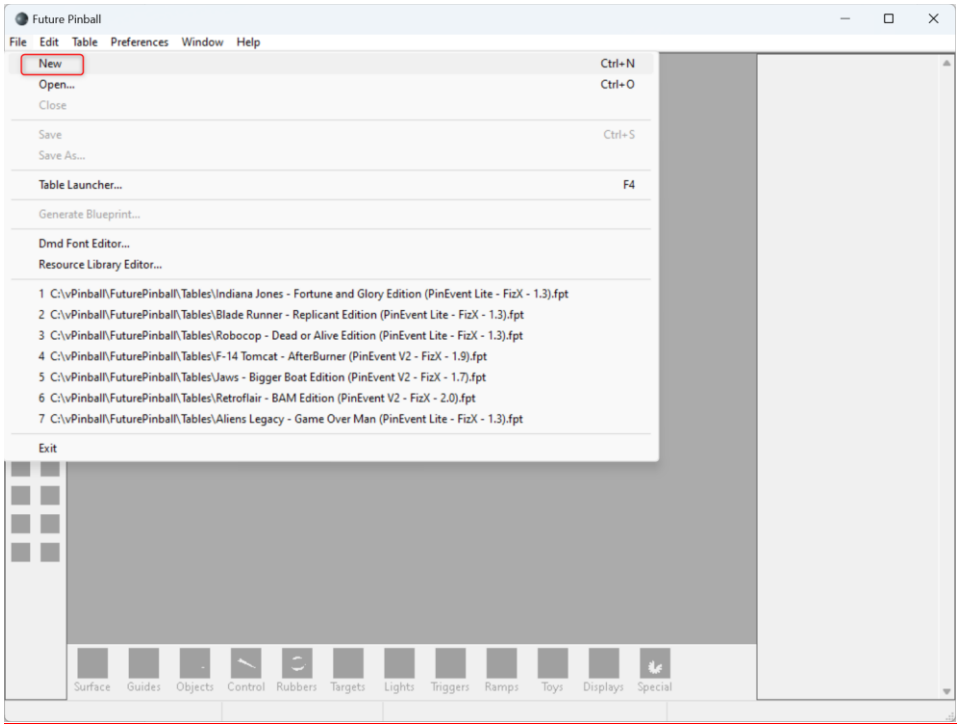
Open Registry Editor and modify following regkeys:

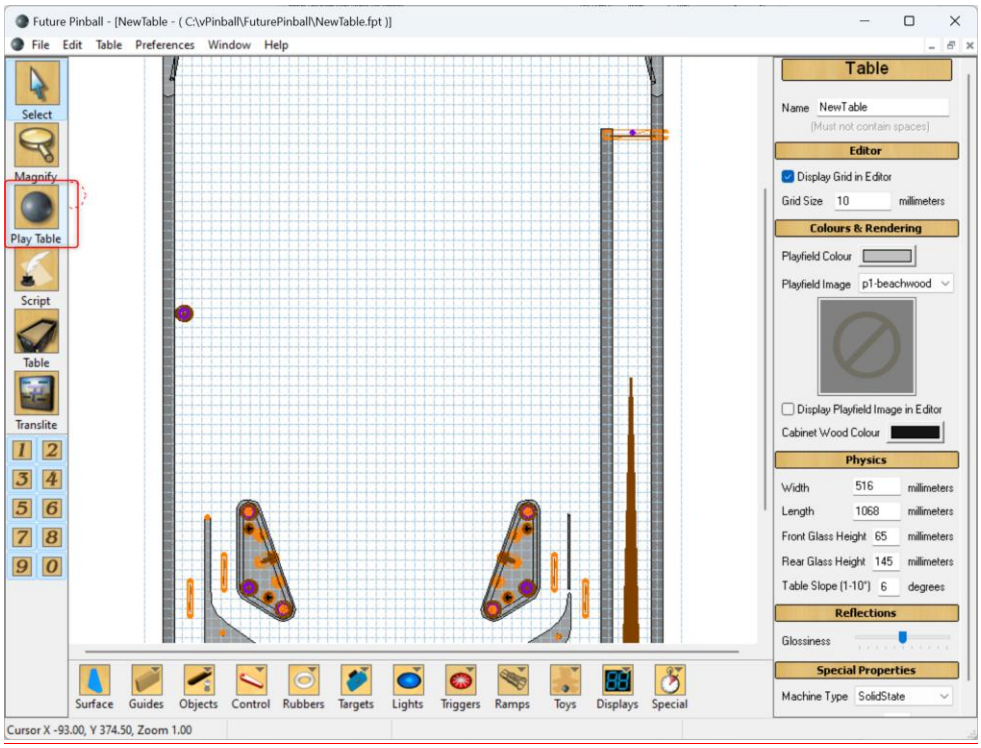


Now configure highlighted (**suggested**) settings as in following pic for analog plunger and nudging:



VERY IMPORTANT NOTE: Only few FP tables support analog plunger; do not waste hours trying it with different tables! To test if it's working, create a new FP Table and play it:

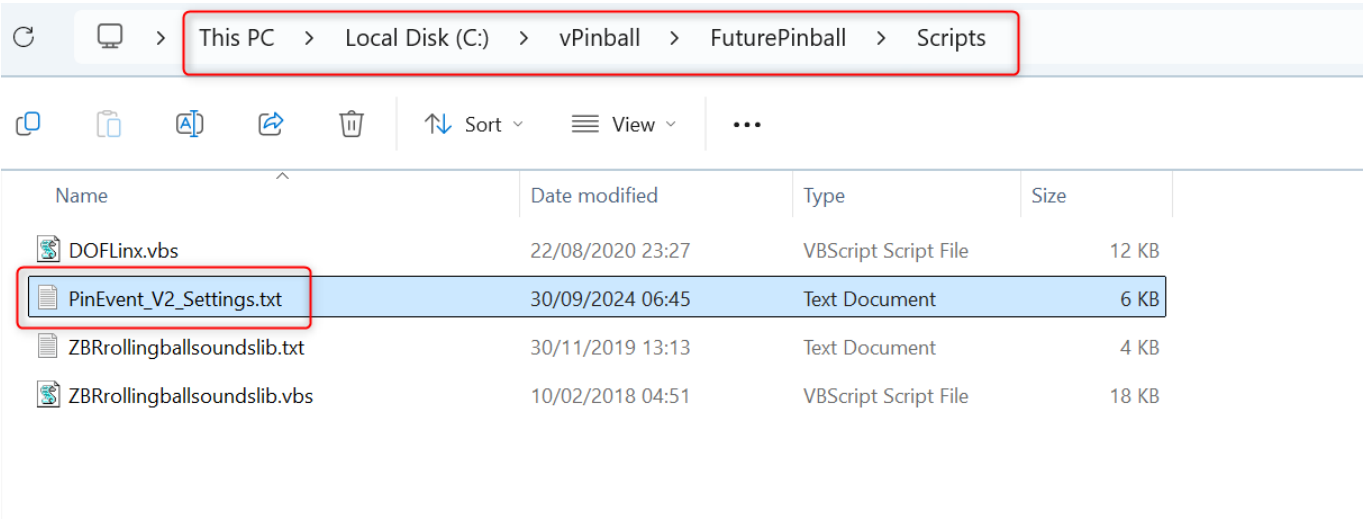




You can now test if analog plunger is working correctly and adjust plunger/nudge parameters in **Game/Keys and Controls** preferences.

DOF Configuration (refer to 5 - PinEvent V2 Guide)

Edit **PinEvent_V2_Settings.txt** file and modify highlighted text:




```
'################ DOF, PUP DMD, PUP SFF - Settings ################

' Change each setting to true for the features you want to use, and for each mode you play in.


'****** Desktop Settings ******

if xBAM.Camera.Mode = 0 then
        DOF_enabled         = true
        PUP_DMD_enabled     = false
        PUP_SSF_enabled     = false
End if

'****** Cabinet Settings ******

if xBAM.Camera.Mode = 1 then
        DOF_enabled         = true
        PUP_DMD_enabled     = true
        PUP_SSF_enabled     = true
End if

'****** VR Settings ******

if xBAM.Camera.Mode = 2 then
        DOF_enabled         = true
        PUP_DMD_enabled     = false
        PUP_SSF_enabled     = false
End if
```

```
'################ NIGHT MODE ################

' When Night Mode is toggled, the DOF noise makers (solenoids, etc) are disabled.

' - Set what key you will use to toggle Night mode
' - Consult the Future Pinball manual to know what the keycodes are. Example: 36 = J key
' - Night Mode Options are:  1 = Future Pinball Sound   2 = PUP SSF Sound

NightMode_Key_enabled = true
NightMode_Key = 36
NightMode_Option = 1
```

```
'################ DOF Settings ################

' Set what will be used for DOF (Direct Output Framework) mechanical feedback for pinball cabinets.

Flipper_DOF_enabled = true
Slingshot_DOF_enabled = true
Bumper_DOF_enabled = true
Solenoid_DOF_enabled = true
DropTarget_DOF_enabled = true
DropTargetReset_DOF_enabled = true
Target_DOF_enabled = true
Knocker_DOF_enabled = true
Chime_DOF_enabled = true
Bell_DOF_enabled = true
Shaker_DOF_enabled = true
Gear_DOF_enabled = true
Fan_DOF_enabled = true
```

IMPORTANT NOTE:

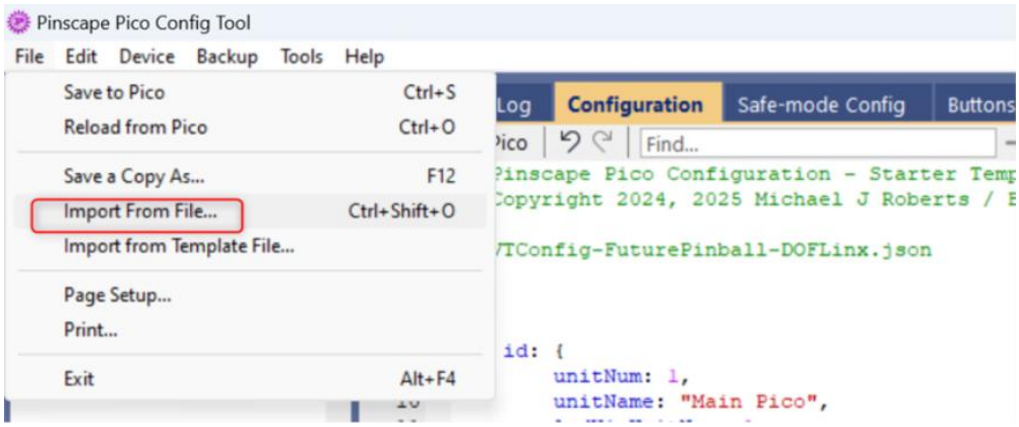
DOF may not work in Future Pinball if you have double registration of the x86 and x64 DirectOutput object (by the RegisterDirectOutputComObject.exe). Follow these instructions to solve:
https://www.vpforums.org/index.php?showtopic=51540

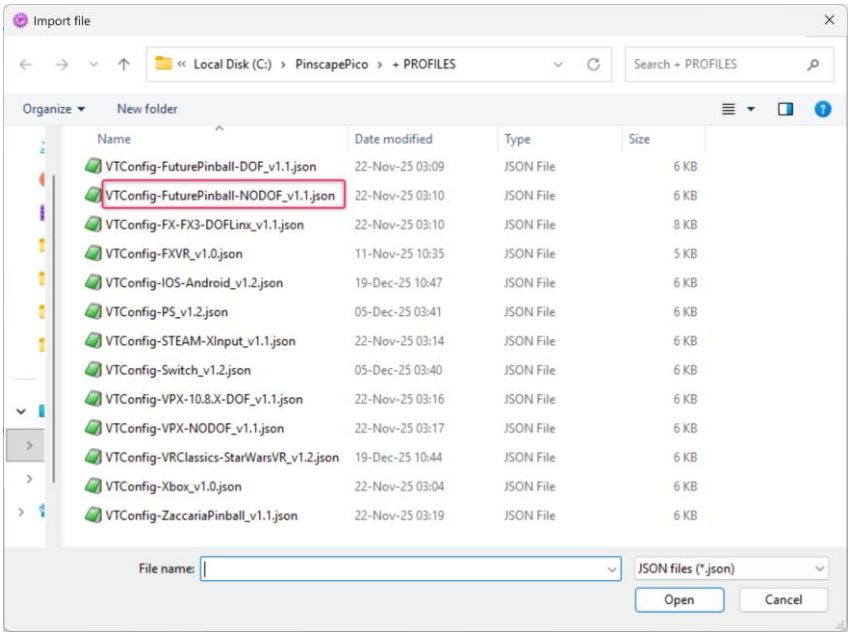
Congratulations! If you follow guides and these instructions, you should now be able to play Future Pinball with full feedback, nudging and analog plunger.

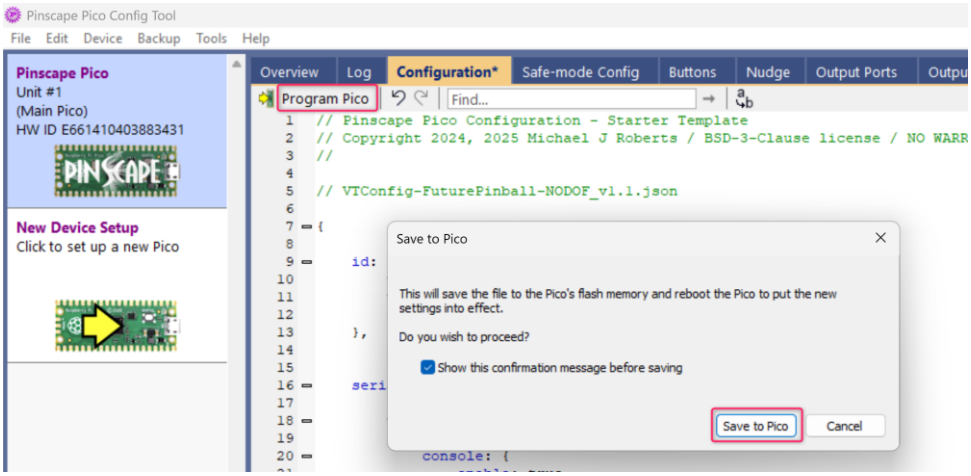
Play Future Pinball without DOF

The new upgraded VirtuaTilt allows you to enjoy feedback from flipper's solenoids and shaker motor even without DOF configured!

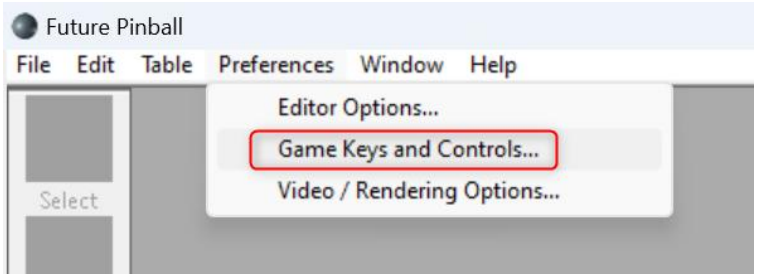
Load **VTConfig-FuturePinball-NODOF_v1.X.json** in **Pinscape Pico Config Tool:**

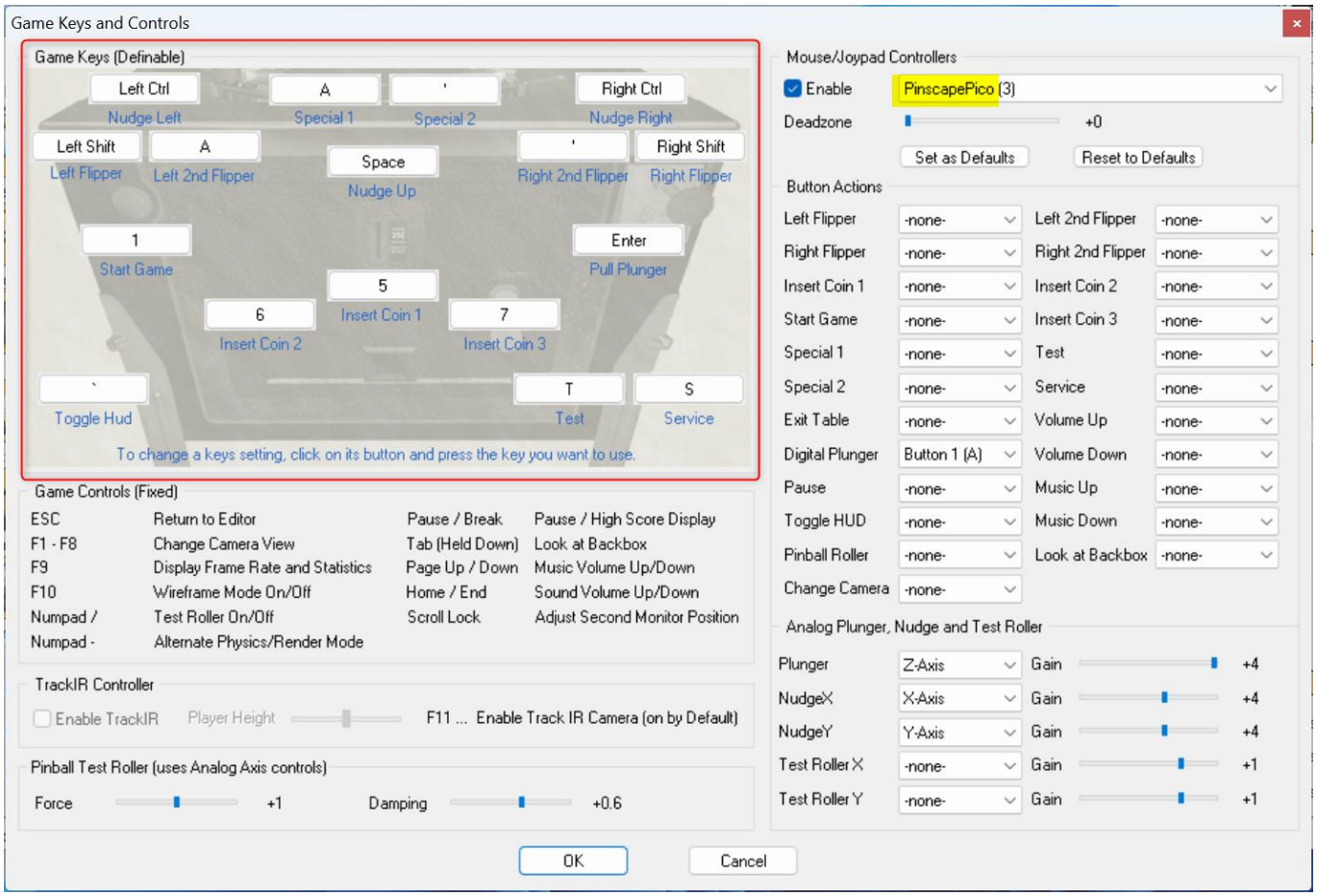






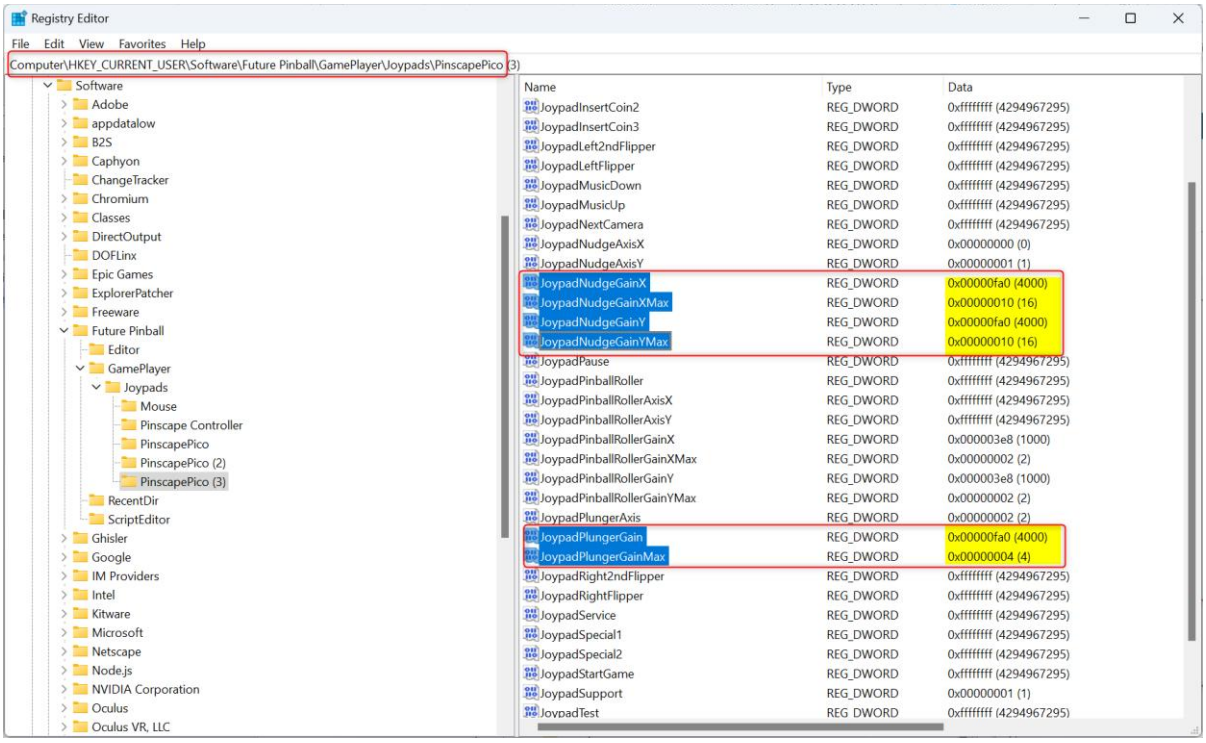
Assign all buttons you need in Future Pinball:



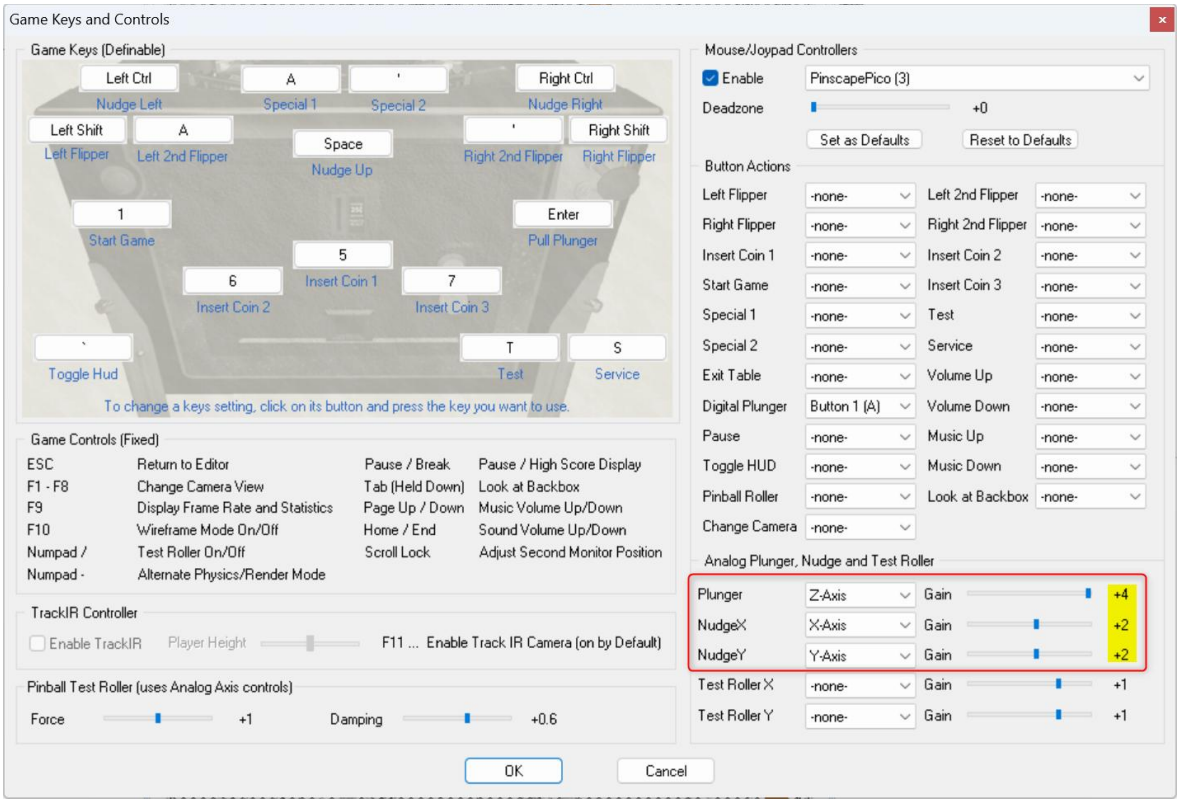


Accelerometer/Analog Plunger configuration

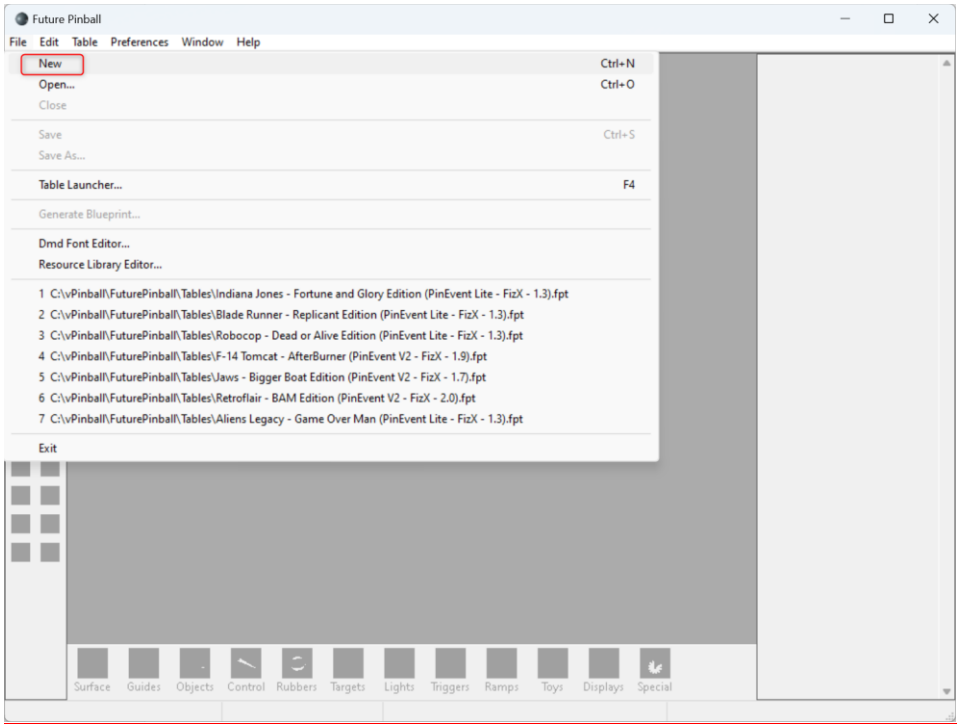
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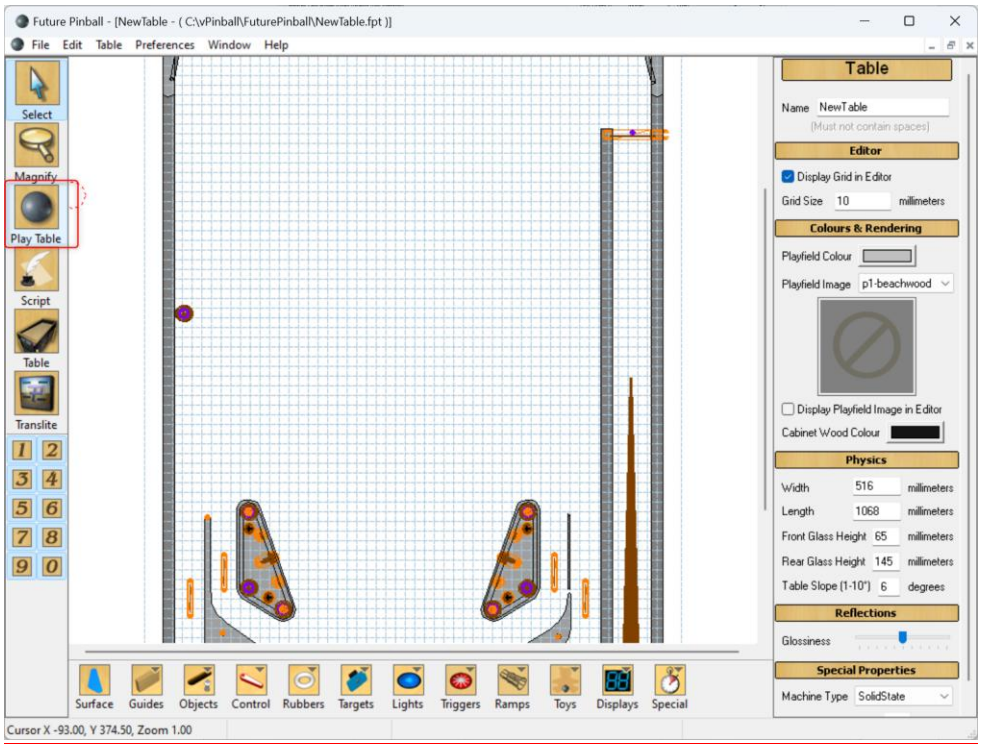


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You can now test if analog plunger is working correctly and adjust plunger/nudge parameters in **Game/Keys and Controls** preferences.

Enjoy!